

TEST BANK

CHAPTER 1

INTRODUCTION TO THE MARINE ENVIRONMENT

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A, and place the corresponding letter from column B in the space provided to the left of term or phrase column A.

[Note to instructors: Currently, each of the choices is correctly matched, so the columns should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. heat of vaporization	—————>	A. hydrogen bonds
___ 2. plate movement	—————>	B. seafloor spreading
___ 3. Leibig's law	—————>	C. limiting factors
___ 4. extinct submarine volcanoes	—————>	D. seamounts
___ 5. copepods	—————>	E. microscopic marine herbivores
___ 6. Coriolis effect	—————>	F. Ekman spiral
___ 7. bacteria	—————>	G. decomposer
___ 8. planktotrophic larvae	—————>	H. larvae stage that feeds on plankton
___ 9. lecithotrophic larvae	—————>	I. larval stage with abundant yolk
___ 10. pycnocline	—————>	J. zone of rapid density change
___ 11. pheromone	—————>	K. a secreted chemical attractant
___ 12. nonpelagic larvae	—————>	L. larval stage with direct development

Multiple Choice

Circle the choice that best completes or answers each of the given questions.

[N.B.: The correct answer to each of the following questions is indicated by an asterisk. For most of the questions, the letter "a" is the correct choice; reorder the choices as you see fit.]

- | | |
|---|-------------------------------|
| 1. Seawater contains about ____ % dissolved substances. | 2. Seawater is normally ____. |
| a. 35 | a. alkaline * |
| b. 3.5 * | b. acidic |
| c. 0.35 | c. neutral |
| d. none of the above | |

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3. The pH range of seawater is approximately _____.
 - a. 3.5 to 5.5
 - b. 5.5 to 7.5
 - c. 7.5 to 8.5 *
 - d. 8.5 to 9.5
4. The pycnocline is the depth zone of which of the following?
 - a. rapid density change *
 - b. minimum temperature change
 - c. maximum temperature change
5. A chain of volcanic, oceanic islands all about the same age, having the same level of erosion, and making a slight arc would probably mean that _____.
 - a. two oceanic plates collided *
 - b. two continental plates collided
 - c. there was a hot spot below the ocean plate at that point
 - d. you wrecked your car
 - e. none of the above
6. Pure water absorbs which of the following colors of light the most rapidly?
 - a. Red *
 - b. Blue
 - c. Green
 - d. Yellow
 - e. Ultraviolet (UV)
7. The water depth of the world's oceans averages about _____.
 - a. 3.8 km *
 - b. 38 km
 - c. 3.8 miles
 - d. 38 miles
 - e. none of the above
8. Approximately ____ % of the surface of the Earth is covered by salt water.
 - a. 20
 - b. 40
 - c. 60
 - d. 70 *
 - e. 80
9. ____ chemical bonds called ____ bonds form between the positively charged hydrogens and negatively charged oxygens of adjacent water molecules.
 - a. Weak, hydrogen *
 - b. Strong, hydrogen
 - c. Weak, covalent
 - d. Strong, covalent
 - e. Strong, nonpolar
10. Water tends to stick to itself because of its _____.
 - a. ability to form hydrogen bonds *
 - b. polar nature
 - c. nonpolar nature
 - d. affinity for nonpolar molecules
11. Pure water is at its densest at which one of the following temperatures?
 - a. 0°C
 - b. 32°F
 - c. 4°C *
 - d. 50°C
 - e. 100°C
12. The oxygen minimum zone generally falls to its lowest value at which one of the following depths?
 - a. 0–100 m
 - b. 200–1,000 m *
 - c. 1,000–2,000 m
 - d. >2 km
13. Among the following plates, which is the single largest?
 - a. Pacific *
 - b. Nazca
 - c. Caribbean
 - d. Iran
 - e. South America
14. Along the edges of most continents are shallow undersea extensions of the land called the ____, which gently grade(s) from the shore to depths of several hundred meters.
 - a. continental shelf *
 - b. continental slope
 - c. abyssal plain
 - d. submarine ridges
 - e. trenches

15. At the outer edge of the continental shelf, the bottom abruptly drops off. This region is called the ____.
- continental shelf
 - continental slope *
 - abyssal plain
 - submarine ridge
 - trench
16. The deep, flat bottom of the ocean floor is known as the ____.
- abyssal plain *
 - continental slope
 - continental shelf
 - submarine ridge
 - trench
17. Tom and Sally are trying to play catch on a merry-go-round that is rotating counterclockwise. To these two, the path of the ball will appear to ____ when they toss it.
- bend to the right *
 - bend to the left
 - fly perfectly straight
18. Because of the Coriolis effect, a water current moving in the northern hemisphere will tend to ____ as it moves along its course.
- curve to the right *
 - curve to the left
 - continue on a straight path
19. When water evaporates from the ocean, the water vapor is most like which of the following?
- Distilled water *
 - Salty ocean water
 - Water that is saltier than ocean water
20. Organisms that capture light energy and store it in the form of the chemical bonds of organic materials are referred to as ____.
- autotrophs *
 - heterotrophs
 - decomposers
 - carnivores
 - herbivores
21. Organisms that consume plants as food are referred to as ____.
- herbivores *
 - carnivores
 - autotrophs
 - decomposers
 - none of the above
22. Which one of the following terms refers to the total amount of living material in a trophic level at any instant in time?
- Standing crop *
 - Trophic structure
 - Autotroph
 - Heterotroph
 - Productivity
23. Boundary zones between adjacent communities are known as ____.
- ecotones *
 - habitats
 - seres
 - dominants
 - niches
24. The change in composition of species within a community is referred to as ____.
- ecological succession *
 - climax
 - limiting factor
 - inhibition
 - species diversity
25. The final, persistent community achieved in a certain geographic region is referred to as the ____ community, while any intermediate communities are called ____ communities.
- climax, sere *
 - sere, climax
 - climax, succession
 - sere, succession
 - limiting, climax
26. Free-swimming larvae that must feed because they possess little yolk are referred to as ____.
- planktotrophic *
 - lecithotrophic
 - nonpelagic

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27. Nonswimming, nonfeeding larvae that possess ample yolk are referred to as ____.
- lecithotrophic
 - planktotrophic
 - nonpelagic *
28. Phytoplankton compose the ____ trophic level of marine food webs.
- first *
 - second
 - third
 - fourth
 - fifth
29. ____ is the term used to describe regions where one plate dives beneath another.
- Subduction*
 - Seafloor spreading
 - Volcanism
 - Upwelling
30. Waves may be generated by which of the following?
- wind
 - earthquakes
 - volcanic explosions
 - all of the above *
31. The ____ is the major north-to-south current in the northern Pacific Ocean.
- California current *
 - Agulhas current
 - Azores current
 - Equatorial countercurrent
 - North Atlantic current (Gulf Stream)
32. The process by which deep water is brought to the surface is called ____.
- upwelling *
 - spring tide
 - fetch
 - benthic flush
 - downwelling
33. Species possessing short life spans, rapid development to reproduction, many reproductive periods per year, and high death rates are called ____ species.
- r*-selected *
 - K*-selected
 - pelagic
 - benthic
34. The model called ____ describes how communities change in an orderly process through modification of the environment.
- inhibition
 - facilitation *
 - tolerance
35. In the ____ model, any species may colonize a site, but succession proceeds when new species that are more tolerant of conditions or are more competitively superior move in.
- inhibition
 - facilitation
 - tolerance *
36. In the ____ model, species are thought not to be competitively superior to one another, and succession proceeds according to which species occupies a site first and then holds it.
- inhibition *
 - facilitation
 - tolerance
37. Organisms that are free swimming and live in the well-lighted region beyond the continental shelf are said to live in the ____ zone.
- epipelagic *
 - mesopelagic
 - bathypelagic
 - abyssalpelagic
38. Benthic organisms that live in the trenches of the world's oceans are said to live in the ____ zone.
- hadal *
 - abyssal
 - bathypelagic
 - epipelagic
39. The very long mountainous regions in the ocean floor, often running down the center of the world's oceans, are called the ____.
- mid-ocean ridges *
 - trenches
 - seamounts
 - volcanoes

40. Atmospheric wind circulation is described by ____ wind belts in the northern hemisphere and ____ in the southern atmosphere.
- three, three *
 - three, two
 - three, one
 - one, three
 - two, three
41. Underwater waves that propagate along the pycnocline are called ____.
- internal waves *
 - surface waves
 - tidal waves
 - Ekman spiral waves
 - none of the above
42. Which one of the following is not a trophic pyramid?
- vitamin pyramid *
 - biomass pyramid
 - energy pyramid
 - number pyramid
43. In a 1,000-g sample of seawater, dissolved compounds compose about ____.
- 35 g *
 - 3.5 g
 - 0.35 g
 - 35 mg
 - 35 μ g
44. At about ____ of depth, the ocean bottom levels off into a flat, extensive sediment-covered plain.
- 4–5 km *
 - 1–2 km
 - 500–1,000 m
 - 100–500 m
 - 1,000 feet
45. Trenches have depths that lie in the range of about ____.
- 10,000 m *
 - 3,500 m
 - 3.5 km
 - 1,000 km
 - 1,000 feet
46. Most marine food chains have about ____ links, whereas terrestrial chains average about ____ links.
- five, three *
 - three, five
 - five, two
 - two, five
 - seven, eleven
47. The continental shelf of the world's oceans covers about ____ % of the total oceanic area.
- 7–8 *
 - 15–18
 - 22–25
 - 30
 - 50

True/False

Mark each of the following statements with the letter *T* if it is true and *F* if it is false.

[Note to instructors: In some cases, a few options are included for the same question.]

- [True] Water evaporates slowly because it possesses a high heat of vaporization.
- [True] Certain bacteria can live in thermal hot springs at temperatures of 85°C.
- [False] Water evaporates slowly because it possesses a high latent heat of fusion.
- [True] When ice is formed, large amounts of heat are given up to the environment because water possesses a high latent heat of fusion.
- [False] When ice is formed, large amounts of heat are given up to the environment because water possesses a high heat capacity.
- [False] The number of major plates composing the Earth's crust is 36.
- [True] Continental crust is less dense than oceanic crust.
- [True] Animals with the ability to regulate their own body temperatures are said to be endothermic.
- [False] Animals lacking the ability to regulate their own body temperatures are said to be endothermic.
- [True] Because of its salt content, seawater freezes at a temperature slightly below 0°C.

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11. [True] The sky is blue because it contains particles that scatter blue light, but allow other colors through; the blue light is then backscattered to the earth.
12. [True] Two of the six inorganic ions that make up over 99% by weight of solid matter in seawater are chloride and sodium. (*substitute two of the following: sulfur / magnesium / calcium / potassium*)
13. [False] Two of the six inorganic ions that make up over 99% by weight of solid matter in seawater are phosphate and nitrate. (*substitute two of the following: silicon / bicarbonate / bromide / strontium*)
14. [False] Terrestrial food chains (webs) usually have more links than marine food chains.
15. [True] On the basis of the ocean's temperature at the surface, four major biogeographic zones have been established.
16. [False] Based on the numbers of phyla or classes present, terrestrial ecosystems are considered more diverse than marine ecosystems.
17. [True] Seamounts are extinct submarine volcanoes that do not reach the surface of the sea.
18. [False] Hydrothermal vents are extinct submarine volcanoes that do not reach the surface of the sea.
19. [False] Seamounts are localized regions on the ocean floor where very hot water that is saturated with minerals wells up from the crust.

Short Answer (Fill in the Blank)

1. Water is a polar molecule because it has a slightly negative charge at the oxygen end of the molecule and a slightly positive charge at the hydrogens.
2. The ability of water molecules to stick to one another is known as cohesion.
3. Water molecules tend to stick to themselves so strongly that small objects may be supported on the surface; this phenomenon is called surface tension.
4. The height that waves reach in the open ocean depends on wind speed, the fetch over which the wind blows, and the duration that the wind blows.
5. The resistance to flow or movement by water is a measure called viscosity.
6. The linkage between the surface currents and the deeper water masses is effected by a process called thermohaline circulation.
7. When carbon dioxide dissolves in water, it forms carbonic acid. [or H_2CO_3]
8. An important chemical property of water is its unusual capacity to dissolve many substances; in this way, it acts almost as a universal solvent.
9. The major ocean currents do not flow parallel to wind direction; rather, they are deflected into roughly circular systems called gyres.
10. Salinity is the term used to indicate the total amount of dissolved material in seawater.
11. The salinity of seawater is about 32 psu.
12. The worldwide circulation of water that runs from the Atlantic Ocean to the Indian Ocean to the Pacific Ocean is called the Global Conveyor Belt (Great Ocean Conveyor Belt).
13. Individual submarine mountains that rise from the abyssal plain, but do not reach the sea surface, are known as seamounts.
14. A dominant geological coastal formation along the Gulf and southeastern coasts of the United States is a series of islands that separate the shallower bays from the Gulf and the Atlantic Ocean, respectively. These formations are called barrier islands.
15. Regions in the abyssal plain cut by deep, narrow troughs are called trenches.
16. El Niño is the name given to an event recognized by a marked decrease in production of algal and fish biomass along the coast of Peru.
17. The four major biogeographical zones are the polar, cold temperate, warm temperate, and tropical [or equatorial] zones.

18. Moving water will impart motion to the layer directly below it; however, because of the Coriolis effect, the motion of each layer is deflected slightly. This circumstance results in a phenomenon called the Ekman spiral.
19. A pathway that transfers energy from plants through a particular series of consumers is called a food chain.
20. The combination of all food chains in a particular community is called a food web.
21. The numerically abundant species within a geographic region are called dominants; they are often used to name or characterize the community.
22. Competition among organisms belonging to the same species is called intraspecific competition.
23. The Earth's crust is divided into a number of large units called plates.
24. The larvae of many sessile, marine invertebrates attach to surfaces near adults of the same species. This phenomenon may occur because the adults secrete a pheromone that serves as a behavioral cue which initiates settling.
25. The dominant marine herbivores are microscopic crustaceans called copepods.
26. The midocean ridges are chains of submarine mountains that stretch for thousands of kilometers down the center of the world's oceans.
27. The movement of two oceanic plates in opposite directions causes seafloor spreading.
28. The movements of the continental and oceanic plates through time is known as continental drift.
29. One region of the world where a continental plate is being torn apart is in Africa.
30. The approximate boundary between cold temperate and polar waters is 50° N or S latitude.

Essay Discussion Questions

Develop complete answers to these questions.

1. What characteristics would you expect to be exhibited by organisms that use the air–water interface as their habitat? What would adding a drop of soap to the water do to them?
2. How would you expect the viscosity (resistance to flow) of water to vary as a function of the size of an organism swimming in it? How does this relationship affect the biology of very small organisms (for example, < 3 mm)? How does this relationship affect the biology of large organisms (for example, > 1 cm or 1 m)?
3. How does the high heat capacity of water affect land near large bodies of water? Give specific examples.
4. How does the latent heat of fusion influence air temperature when it begins to snow? Why do fruit growers in southern states such as Florida and Texas spray water on their crops when a freeze is expected?
5. How does the latent heat of vaporization influence air temperature when it begins to rain?
6. Describe the peculiar relationship between temperature and water. At what temperature (in °C) is water density the greatest? What would the marine environment at the poles be like if the density of water were to increase as the temperature approaches 0°C? Discuss.
7. For health reasons, some physicians suggest reducing your NaCl intake. With this advice in mind, why would the strategy of just substituting sea salt for table salt in one's diet not be a wise idea?
8. Why might you expect to find an oxygen minimum zone (OMZ) below an area of high primary production in oceanic systems? Why might the physical location of the OMZ not be exactly below the region of high productivity? Where would the OMZ be in the water column? Would it be close to the surface, deeper in the sea, or at the very bottom? Discuss the vertical profile of oxygen concentration in the ocean and in lakes. [*Instructors may need to do some of the background work here.*]
9. What physiological activity would you expect to be going on at a relatively high rate in the oxygen minimum zone? What group of organisms would be responsible for this activity? What are the organisms doing?
10. Would you expect CO₂ to be limiting to the photosynthesis of plants in the sea? Why or why not?
11. What is acid rain? What will happen to the pH of distilled water if acid rain lands in it? What will happen to the pH of lake water if acid rain lands in it? Consider both a soft-water lake and a hard-water lake. What will happen to the pH of seawater if acid rain lands in each lake?

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12. If carbon dioxide gas is removed from seawater, the equilibrium will be disturbed and carbonic acid and bicarbonate will shift to the left in the following equations: What will happen to the pH of the system if CO₂ is removed? If bicarbonate is removed? How much would you expect the pH to shift? What activities would remove carbon dioxide from marine water?
13. The continental shelf extends offshore for up to 400 km off eastern Canada, but extends only a few km offshore along most of the Pacific coast of North America. Explain this relationship in terms of plate tectonics.
14. Explain why the sediments on the abyssal plain are thickest as you get closer to the continental slope.
15. Explain why the sediments on the abyssal plain are thicker than on the slopes leading up to the midocean ridges.
16. What does the term ectothermic (poikilothermic) mean? When the water temperature changes, what are the consequences to the physiology of the organism?
17. If ectothermic animals like fish live in seawater, which is actually about 0°C, why don't they freeze? [Hint: Why doesn't your car's radiator freeze in winter?] Discuss.
18. What keeps the various water masses (upper, middle, and deep) from mixing?
19. Does a wave produce much, if any, horizontal movement of water? If not, then what is transmitted horizontally? Taking into account your answers here, describe why a submarine earthquake many kilometers out to sea can cause such damage on land by tsunamis (tidal waves).
20. Why don't waves have much effect in disrupting a thermocline? To what do the terms wavelength, frequency, and period refer in relation to waves?
21. Explain the Coriolis effect. Why is it not considered to be a real force? Discuss.
22. Explain the Ekman spiral. How is it related to the Coriolis effect? Of what significance is it in the open ocean?
23. Describe and explain coastal and equatorial upwelling. Of what significance are these phenomena to coastal fisheries?
24. With few exceptions, the original energy source (for all ecosystems) is the sun. Discuss the exceptions. How do the organisms that live at the lowest level of the food chains in these special ecosystems obtain energy?
25. What is the efficiency of energy exchange between trophic levels? Between what two types of trophic levels would you expect the transfer to be most efficient? The least efficient?
26. What is the significance of eliminating a single species from a food chain or food web? Are such extirpations always significant in terms of the population dynamics of the other members of the ecosystem?
27. What is meant by the term biogeochemical cycle? Provide examples.
28. Explain Leibig's law of the minimum in terms of a nutrient that is essential to the growth of an alga.
29. Name some marine ecotones. What are the boundaries in each of these ecotones?
30. In general terms, what is a community? Compare the concept of community in terrestrial and oceanic environments.
31. Name a species that is limited by one (or more) of the following factors: light, food, nutrients, water, space, and predators. Discuss how it is limited.
32. Distinguish between a food chain and a food web. Which is more complex? Provide a simple illustration of each to support your discussion.
33. What is a keystone [key industry] species? What is a keystone [key industry] predator? How are humans a keystone [key industry] species?
34. Discuss the concepts of intraspecific and interspecific competition.
35. Define the following terms: competitive exclusion principle, grazer, and filter feeders.
36. How does the viscosity of water influence the dynamics of filter feeding?
37. What sorts of marine communities exist without the benefit of an autotrophic component?

38. Distinguish between species richness, species evenness, and species diversity.
39. What mechanisms do planktonic organisms use to stay afloat?
40. Discuss the kinds of skeletons marine invertebrates possess. [*Pardon the following teleological phrase.*] If marine organisms are buoyed by water so that they do not need to have a significant amount of their biomass invested in structural material such as skeletons or tests in order to hold themselves erect against the force of gravity, why do they have these skeletons?
41. Among terrestrial organisms, the dominant compounds are carbohydrates, especially cellulose; among marine organisms, the predominant material is protein. What effect does this difference have on the sediments in marine systems as compared with sediments in terrestrial systems?
42. How can a pycnocline or thermocline represent an ecotonal boundary? Discuss.
43. Among terrestrial organisms, the dominant compounds are carbohydrates, especially cellulose; among marine organisms, the predominant material is protein. Discuss the decomposition of these two types of materials. How will the pH change when these two types of materials are decomposed?
44. What are the principal carbohydrates of terrestrial ecosystems? Of marine ecosystems? What is their basic biochemical structure?
45. What is a copepod? What is the genus name of the most abundant copepod in the sea? How are copepods important to humans?
46. Copepods and other small herbivores of the sea usually consume entire algal cells. Can an algal cell be swallowed and “live to tell about it”? What benefits might such a capability have for the alga?
47. Compare and contrast the relative number of trophic levels in marine and terrestrial ecosystems. Compare and contrast the relative primary productivity of marine and terrestrial ecosystems. Discuss how much harvestable protein is available from each type of system.
48. The production of organic material is usually higher in terrestrial ecosystems than in marine ecosystems, but why is the efficiency of transfer from the first to the second trophic level higher in marine ecosystems? Discuss.
49. Evaluate and discuss the efficiency of the energy transfer from the first to the second, and then from the second to the third, trophic level in terrestrial and marine ecosystems.
50. Why do surface oceanic water temperatures vary less from season to season than air temperatures over nearby land masses?
51. Discuss the processes that occur when ice forms on seawater. Explain why these processes, once begun, tend to establish conditions that resist further freezing. Does salt water freeze in the same way as distilled water?
52. How are the surface currents of the North Atlantic similar to those of the South Atlantic? In what major ways do they differ?
53. A ship loses some of its cargo of track shoes overboard during a heavy storm off the east coast of New Zealand. Assuming that these shoes float for a long time, where might they end up? Discuss several possibilities.
54. List and describe several major physical and chemical features of seawater that change markedly from the sea surface downward. How do these same features change along the sea surface as one proceeds from the equator north or south to higher latitudes?
55. Discuss the general principle involved in the following concept: Marine food chains tend to be much longer than terrestrial food chains, although both may end in animals of similar size (or at least a size of about the same order of magnitude).
56. Provide examples of and discuss the differences between the life histories of *r*-selected and *K*-selected species.
57. Discuss the different number of phyla present in terrestrial and marine ecosystems.
58. On average, terrestrial ecosystems have fewer links in their food chains than do marine ecosystems. Discuss why one system contains more phyla than the other.
59. How are deep ocean currents driven by temperature and salinity differences?

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60. Draw a vertical profile depicting the relative concentrations of oxygen from the surface to a depth of 2,000 m in the open ocean. Label the region of the oxygen minimum zone, and explain why it develops there.
61. Why do waves increase in height and finally break when they enter shallow water?
62. Discuss the differences between the following terms: planktotrophic, lecithotrophic, and nonpelagic.
63. Compare and contrast *r*- and *K*-selected species. Give examples of each from the marine and terrestrial worlds.
64. List five environmental factors (cues) that appear to be important to larval settlement and metamorphosis, and discuss how the larvae use two of these cues to enhance their own survival.
65. Discuss and present supporting examples of the two life history strategies known as the *r*-selected and *K*-selected strategies. By what other names are these strategies known?
66. Gravity is less important in marine ecosystems than in terrestrial systems. How does this factor influence the amount of structural material and stored energy in marine vs. terrestrial organisms?
67. Sound travels faster in water than in air. How does this factor influence marine and terrestrial predators that use sound to detect prey?
68. Develop an argument that explains why large marine mammals occupy higher trophic levels than do large terrestrial animals.
69. Describe the concept of competitive interference (interference competition), and compare it with exploitive competition.
70. In some warm, shallow seas, the water takes on a milky-white tint when viewed from the air during times of intense algal production. Explain why.
71. Compare and contrast the facilitation, inhibition, and tolerance models.
72. You are in the southern hemisphere at 45°S latitude facing north and shoot a harmless rocket aimed north. Where will the rocket land? (Assume no interference by winds, birds, flying saucers, or other apparitions.)
73. The Coriolis effect causes moving air masses to curve to the right while they move over the surface of the Earth in the northern hemisphere and to curve to the left in the southern hemisphere. Explain this concept to someone who is not familiar with it by using the analogous situation of a merry-go-round.
74. Why wouldn't waves having a wavelength of 60 meters be noticeable to a scuba diver at a depth of 40 meters?
75. How has human culture been affected by tsunamis?
76. Why is the western coast of the United States less susceptible to damage by tsunamis than the coastlines of other Pacific countries?
77. What are the three larval types that Vance (1973) described? What are the advantages of each? Discuss.
78. How would you expect each of the larval types described by Vance (1973) to be cued (in their production) to the primary production of the shallow-water zone?
79. Which of the three larval types described by Vance (1973) would you expect to give the most genetically diverse adult population in any one area? The least genetically diverse? Why? Discuss.
80. What sorts of cue(s) do larval stages use to test a substrate to determine whether it is suitable as a permanent habitat? Why would larvae use chemicals associated with adults of the same species as a cue? (This question is somewhat teleological, but answer it in evolutionary terms.)
81. Define the following terms and provide an example of how each of them might be used: pheromone, allomone, kairomone, and allelochemic.
82. Consider the adaptive significance (in evolutionary terms) of a larval stage that will be absolutely specific to one particular substrate and that would die if that substrate were not present. Compare this larval type with one that will accept substrates other than its preferred one.
83. What physical or chemical factors besides the ones discussed in question 80 are important to larvae as cues for settling or not settling?

84. What one or more ecological forces play an important role in determining where *r*-selected species are found? *K*-selected species? Discuss.
85. *K*-selected species appear to be better competitors once they are established in a particular habitat. What can happen to upset the balance in favor of *r*-selected species once *K*-selected species have become established?
86. Develop an argument that would explain why, in any given habitat, there would probably never be just *r*-selected or just *K*-selected organisms. What sorts of habitats would be inhabited mainly by *r*-selected species? by *K*-selected species?
87. Why would a protozoan be considered to be a *r*-selected species? Discuss.
88. Why would a whale be considered to be a *K*-selected species? Discuss.
89. Why are there no terrestrial animals with the ability to detect electrical activity in other animals, as is the case in marine animals? Discuss.
90. Many grazers in terrestrial ecosystems (e.g., bison, antelope, horse, and termites) have guts loaded with symbiotic microorganisms. This phenomenon is not seen in marine grazers such as copepods. Why might this be the case?
91. Other than perhaps spiders, why are there no terrestrial filter feeders?
92. Compare the concepts "species richness" and "species diversity" in terms of the amount of information each provides.
93. Construct a simple food pyramid composed of the following organisms: single-celled algae, bacteria, herrings, squids, and copepods.
94. Compare biological diversity in the world's oceans with that on land. Which region is more diverse? Speculate on the reason(s) for these differences.
95. Speculate on what would happen if the Great Oceanic Conveyor Belt were to stop abruptly.
96. Challenge Question: Determine the oxidation states of the nitrogen atom in the following ions: ammonium, nitrite, and nitrate. Of what significance is this information?
97. Challenge Question: Given the Coriolis effect, why do the winds of a hurricane cycle in a counterclockwise direction in the northern hemisphere?
98. Challenge Question: The biogeochemical cycle of one element is currently way out of balance in the world. Which element is that? Why is the cycle so out of balance for this element?

